FTEC5660 REPRODUCIBILITY WORK

A-MEM Memory Evolution Ablation

Testing full A-MEM against a link-only variant on LoCoMo

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The paper claims that Memory Evolution improves A-MEM

Multi-Hop F1 from the paper

27.02
Full A-MEM

21.35
w/o ME

3 seeds
tested locally

A-MEM pipeline

01 **Note** — 02 **Retrieve** — 03 **Link** — 04 **Evolve**

Scope: official benchmark code, official LoCoMo data, DeepSeek Chat, one conversation, 25 questions.

One isolated change: remove neighbor updates

FULL A-MEM

1. Builds structured memory notes
2. Retrieves similar memories
3. Creates links between related notes
4. Updates neighbor context and tags

LINK-ONLY / w/o ME

1. Uses the same note construction
2. Uses the same retrieval and link step
3. Disables neighbor context updates
4. Disables neighbor tag updates

The ablation is isolated and measurable: only Memory Evolution changes.

Overall mean F1

41.76
Full A-MEM

38.25
Link-only

+3.51
F1 points

Paper vs this run

Target metric	Paper GPT-4o-mini	This run DeepSeek
Multi-Hop full	27.02	44.24
Multi-Hop w/o ME	21.35	28.69
Temporal full	45.85	11.59
Temporal w/o ME	31.24	12.62

Multi-Hop direction reproduced. Temporal did not.

The pipeline worked. The paper's broad advantage did not.

WORKED

- Six runs completed
- 25 questions per run
- Zero LLM failures
- Multi-Hop mean gain +15.56

DID NOT HOLD

- Temporal mean change -1.03
- Single-Hop mean change -11.43
- Large seed-to-seed variance
- Small sample makes results uncertain

KEY LESSON	Match model, data version, scoring, and retrieval k before comparing scores.
NEXT STEP	Start with a small test, verify failures and caches, then expand questions and seeds.
CONCLUSION	**Partial behavioral reproduction, not an exact numerical replication.**